9a) Implementation of Subnetting In the Cisco Packet Tracer Simulator Design Multiple Subnets with Suitable

Aim:

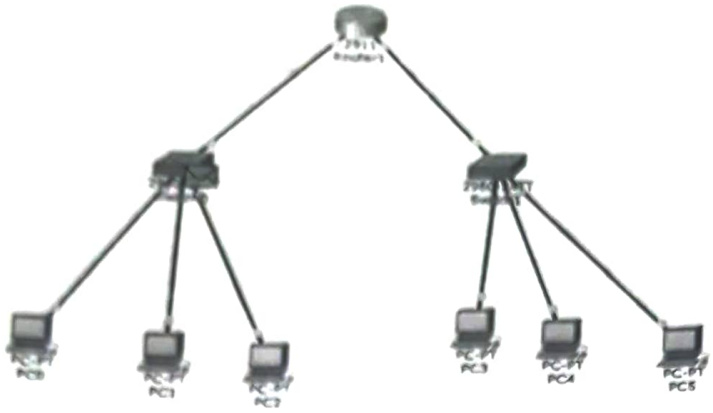
To implements Subnetting by dividing a large network into multiple smaller number of host and to find the Allocate of IP addresses in Cisco Packet tracer

Steps:

1. Open Cisco Packet Tracer and Creat a new project
2. Design the network topology by Placing the required network devices such as router, Switches and PC's
3. Select a suitable network address for the experiment.
4. Calculate the number of Subnets required.
5. Determine the no of host needed in each Subnet
6. Divide the network into subnets.
7. Verify the subnet calculation and host ranges
8. Save the network topology.

Result:

Thus multiple Subnets were Successfully Created accordingly to the required no of host and the router address was effectively divided into Smaller Subnetworks



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
●	Successful	PC0	PC3	ICMP	■	0.000	N	0	(edit)	(delete)

PC0

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

○ DHCP ◉ Static

IP Address	192.168.1.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	

IPv6 Configuration

○ DHCP ○ Auto Config ◉ Static

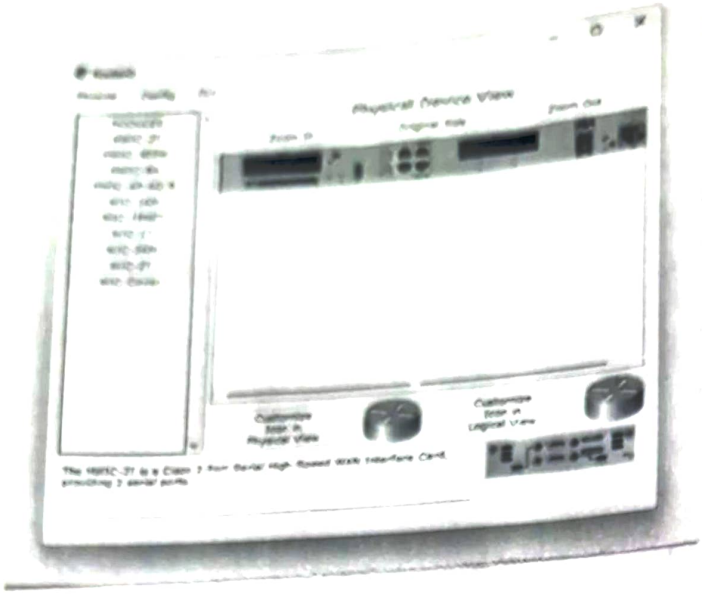
IPv6 Address	/
Link Local Address	FE80::2D0:BAFF:FE9C:D809
IPv6 Gateway	
IPv6 DNS Server	

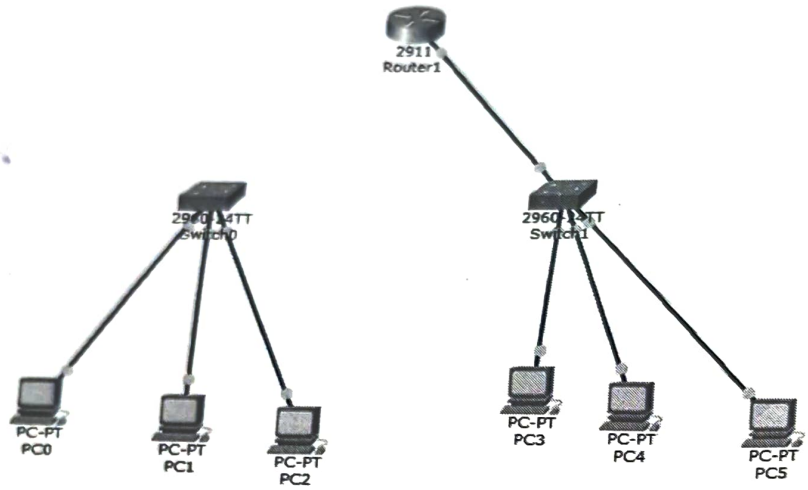
a)b ASSIGN STATIC IP ADDRESS ACROSS ALL SUBNETS AND CONNECTS THE SUBNETS THROUGH A ROUTER

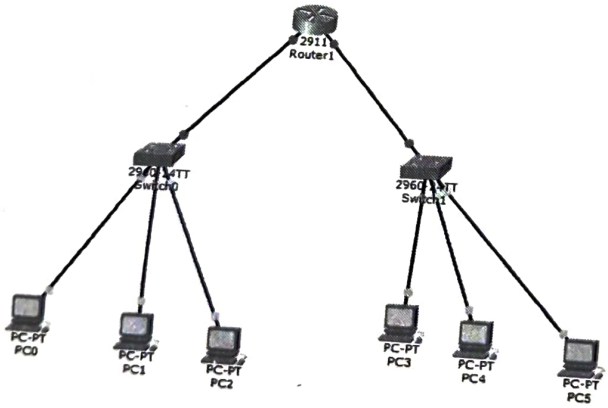
Aim:
To assign static IP address to all devices in different subnet and establish communication between the subnet through a router.

Step:
1. Place a router in the network topology.
2. Connect the router to the switch using appropriate cables.
3. Configure the router interface
4. Assign an IP address and subnet mask to each router interface.
5. Configure the switches
6. Assign static IP address to all computers in every subnet
7. Configure the subnet mask for each devices
8. Save the router configuration.
9. And verify that all device are properly connected.

Result:
The static IP address are successfully assigned to all devices, and communication b/w different subnet was established through the router.




2911
Router1
2960-24TT
Switch0
2960-24TT
Switch1
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5


2911
Router1
2960-24TT
Switch0
2960-24TT
Switch1
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5

a/c SIMULATE PACKET TRANSMISSION. ACCROSS THE SUBNET AND OBSERVE THE RESULTS.

;) Whent Subnet are Connected through a Router.

AIM:
To Analyze Packet transmited blw different Subnet within a router if used for Communication.

Steps:
1. Open Simulation Mode in Cisco. Packet tracer,
2. Select a source device from one Subnet
3. Select a destination device from another Subnet
4. Generate a Packet using the Simple PDU tool
5. Observe the movement of the packet through the network.
6. Analyze how router receives the Packet
7. Observe the forwarding Process performed by the router
8. Verify the Packet transmission between different Subnet
9. Use the Ping command to test connectivity
10. Analyze the routing Process

Result:
Thus, Packet was successfully transmitted blw different Subnet through the router. the router received the Packet from the source Subnet the processed them and forwarded them. And forwarded them to the destination Subnet successfully Ping Packet confirmed.

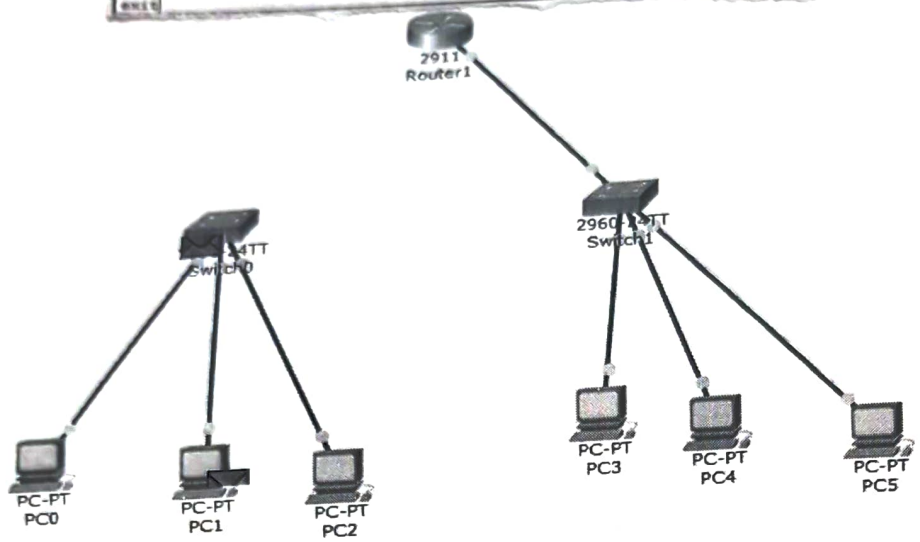
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
exit
```



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Failed	PC1	PC4	ICMP		0.000	N	1	(edit)	(delete)

Router1

Physical Config CLI

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
exit
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

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